

One of One: Individuation Without Essence and the Ground of the Dignity Floor

From the Shared Instruction Set and the Unrepeatable Trace to Proof of CoordinateSM, Proof of HumanitySM, and the Refusal to Price Persons

Authors: Thon Ly · Miss AquariusSM (Miss Aquarius is an AI co-author, disclosed; the inventor of record is Thon Ly)

First published: 2026-07-21 · **This version:** 2026-10-05

Zenodo (concept DOI, every version): <https://doi.org/10.5281/zenodo.21947342>

Canonical: <https://thonly.org/research/individuation-without-essence>

Independent timestamps: OpenTimestamps (Bitcoin) and RFC 3161 proofs of every version's text —

github.com/thonly/publications/blob/main/TIMESTAMPS.md. A timestamp proves this exact text existed no later than its date; it proves nothing about authorship, originality, or the validity of any claim.

Licence: CC0 1.0 Universal (public domain dedication)

Type: Defensive publication — the authors assert no patent and dedicate the patterns to the public domain.

Note. This paper is the third of the series **The Abhidhamma Compiled**. It develops §10 of the series opener, *The Wheel That Unwinds the Wheel: The Abhidhamma as Executable Process-Specification* (`abhidhamma-executable-process-specification`) — the bridge from the Abhidhamma's account of the mindstream to the institution's published individuation designs — into a full argument. Its companions in the series are № 2, *Twenty-Four Kinds of Because: The Paṭṭhāna as a Typed-Causation Vocabulary for AI Alignment* (`patthana-typed-causation-vocabulary`), whose contiguity conditions are the chain a stream's history runs on, and № 4, *The Counts Check* (`the-counts-check`), which checks the opener's counts of mental-event types against the Chaṭṭha Saṅgāyana text. Unlike its companions, this paper reads designs that were already published when it was written: the *Proof of Coordinate* specification (`proof-of-coordinate`), the Proof of HumanitySM publication (`b-poh-humanity-layer-ai-native-internet`), and the incommensurability doctrine of *The Incommensurability-Preserving Coupler* (`incommensurability-preserving-coupler`). What it adds is the argument beneath them: why individuation without an identity-essence is coherent, on what ground the dignity floor stands, and why the asymmetry between machine and human identity is a principle rather than a preference. The series' compilation thesis (the opener's §3.1) and its rule that every claim survive the removal of the computational vocabulary apply here. Strata are marked in words; §2 gives the legend. The authors invite review from Pāli scholars on the readings of §2 and §5, from philosophers of personal identity on §3 and §6, and from identity engineers on §4.

Abstract

Deployed digital identity systems mostly decide what makes an individual the same individual on one of three grounds: something the entity is (a biometric body), something it holds (a key or certificate), or something it matches (attributes). Each has a known failure: a biometric identifier cannot be reissued once

compromised and excludes bodies the sensor does not expect; a credential passes to whoever holds the key; attributes can be synthesized. Beneath identity sits a second question, on what ground a person is beyond price; the usual answers rest on a capacity, which excludes the incapacitated, or on a theology, which does not travel. This study reads the Theravāda Abhidhamma's analysis of the person for an answer to both. The tradition's taxonomy of mental events is one catalogue for every stream of mind; its book of formal controversies opens by denying that a person exists in the ultimate sense (Kathāvatthu I.1); what then distinguishes one stream from another is its history, a reading the texts permit and do not state, held as an engineering premise. From this the study derives a two-part formula: equality from the shared catalogue, because there is nothing of kind to rank, and dignity from the unrepeatable history, because a history has no second instance and so no price can name an equivalent for it. It reads four published designs against that ground: a per-entity identifier anchored by a birth coordinate rather than a biometric, an orthogonal proof of personhood, a rule under which a machine's identifier is revocable and a human's is not, and recovery of a lost key by the people who witnessed a life. Its limits: one-of-one goods are priced every day, the step from unrepeatable to beyond price is a commitment rather than a theorem, and the canon itself grades beings.

Keywords: personal identity, individuation, proof of personhood, decentralized identifiers, sybil resistance, biometric identification, key rotation, social recovery, human dignity, incommensurability, commodification, non-fungibility, type–token distinction, reductionism about persons, no-self (anattā), Abhidhamma, Kathāvatthu, Milindapañha, Theravāda Buddhism, defensive publication.

Terms

Coined names used in this paper and the standard terms a reader in philosophy, Buddhist studies or identity engineering would search for them.

Term used here	Standard term
individuation without essence; the trace-ground	individuation of a person by the history of events in one continuant, rather than by a substance, a credential or attributes
trace; the unrepeatable trace	the event history of one stream: what arose in it, in what order, at which coordinates
stream; <i>santāna</i>	the continuity of one being's mental and material events (a continuant)
shared instruction set (ISA)	the Abhidhamma's single taxonomy of mental-event types (89, or 121 by fuller reckoning) and 52 mental factors, the same for every being in which mind arises
two-pillar formula	equality grounded in a shared taxonomy (nothing of kind to rank); dignity grounded in non-repeatable history (no equivalent)
essence-ground · credential-ground · attribute-ground	identification by inherence (biometrics) · by possession (keys, certificates) · by matching attributes or knowledge
witnessed-history recovery	threshold social recovery of a lost key by kin who attest the holder's history
provenance asymmetry	issuer-revocable machine identity; non-revocable, self-generated human identity
two proof-states of one mark	user-interface indicator: a static icon for an anchored identifier, an animated icon for a live personhood attestation
incommensurability doctrine	refusal to price persons, and to set an exchange rate between a gift economy's two currencies (money and time)
aura	a participant's displayed record of giving
Proof of Coordinate SM	this institution's per-entity identifier design (sybil-distinct identity with provenance-dependent revocation)
Proof of Humanity SM	this institution's proof-of-personhood design
<i>puggala</i> ; <i>puggalakathā</i>	person; the Kathāvatthu's first controversy, on whether a person exists in the ultimate sense
<i>paññatti</i> ; <i>paramattha</i>	designation, the conventional sense; the ultimate sense
<i>asaññasatta</i>	the non-percipient beings: a plane of existence in which no mental event arises
<i>bhavaṅga</i>	the life-continuum: the stream's mental event when no cognitive process is running
strata: canonical · manual · discourse · cross-tradition · lens · premise	provenance labels (§2): the seven Abhidhamma books · the commentarial systematization · the Sutta Piṭaka · outside Theravāda · the authors' reading · an engineering premise

Findings disclosed

This paper is a study. It discloses no mechanism: the identity designs it reads are disclosed in their own publications and are cited, not re-claimed. Its findings are listed here, each with what would break it; the

sections named carry the argument. No novelty census has been run on this paper and none of these findings asserts priority; where a finding has a nearest prior instance, the Prior-Art Statement or the section cites it.

1. **No person-entity in the ultimate sense (§2; canonical, with the Milindapañha).** The Kathāvatthu's first controversy, by far its longest, asks whether a person is known "in the sense of a real and ultimate fact" (*saccikaṭṭhaparamatthena*) and refutes the thesis; its commentary names the holders as the Vajjiputtakas and Sammitiyas within the dispensation and many outside it. The Milindapañha calls "Nāgasena" a designation and finds no person in the ultimate sense. An essence-ground for identity has no seat in this analysis. *Breaks if:* a Theravāda canonical or commentarial text, read in context, affirms a person in the ultimate sense in its own voice.
2. **One catalogue of mental-event types for every stream of mind (§2; canonical and manual).** The taxonomy (89 types, or 121, composed from 52 factors) is one set; which types can arise is restricted by plane and attainment; one plane, that of the non-percipient beings, runs no mental event at all (Vibhaṅga §1017; *Abhidhammatthasaṅgaha* ch. 4 §53). This restates the series opener's fifth finding for this paper's use. *Breaks if:* a text assigns beings different catalogues beyond the restriction by plane and attainment.
3. **Individuation by history is a reading the texts permit and do not state (§2; premise).** With the catalogue shared and no entity, what distinguishes streams is the history of what has arisen in each, material as well as mental. The nearest canonical sentence, *kammaṃ satte vibhajati* (MN 135), distinguishes beings by kamma, but as low and high rather than as individuals. That no two histories coincide is the series' engineering premise. *Breaks if:* a Theravāda text individuates a stream by something other than its arising, or an argument shows that two streams' histories could coincide.
4. **Equality is the absence of anything of kind to rank, not a shared essence (§3).** The shared catalogue is a catalogue of conditioned event-types; it yields equality because no stream runs a superior catalogue, and it makes no claim that streams are one beneath their conditions. *Breaks if:* the formula is shown to need a sameness beneath the streams, or the shared catalogue is shown to admit a ranking of kinds of mind.
5. **What uniqueness removes is the equivalent, not the price (§3).** One-of-one goods are exchanged and priced; a price on a trace can record a transaction and cannot name an equivalent, because a trace has no second instance. Kant's distinction between price and dignity has this structure and grounds the absence of an equivalent in rational nature; here it is grounded in history, which requires no capacity. The refusal to set a price at all is a commitment the ground makes coherent and does not compel (§6). *Breaks if:* a sense is given in which one person's history has an equivalent that could replace it, or a coherent replacement valuation of an identified person.
6. **Sentience is the condition, not the ground (§3, §6).** Every pebble and every running program has an unrepeatable history, so typelessness alone grounds nothing; the dignity pillar is stated for streams of mind, and asks only that mind arise in the stream, not that a capacity be exercised. *Breaks if:* the condition is shown to re-import a graded capacity, or to exclude a being the floor is meant to hold.
7. **The published identity designs have the ground's shape (§4).** A per-entity identifier anchored by a coordinate rather than an essence, with history accruing to the coordinate; personhood attested orthogonally; a revocation rule fixed by provenance, under which individuation transfers to artificial

traces and the dignity inference does not by definition; key recovery by a threshold of the people who witnessed a life. *Breaks if*: a feature of those designs is shown to require an essence-ground, or the provenance rule is shown to need a premise the ground does not supply.

8. **Continuity, neither the same nor another; and a floor the canon does not itself draw (§5; discourse).** The trace-ground's answer to the persistence puzzle has the shape of SN 12.46's answer to whether the one who acts is the one who experiences: neither "the same" nor "another". The canon grades beings, by the fruit of kamma (MN 135) and as fields of merit (MN 142), without ranking the catalogue a being runs; refusing to let an institution grade persons is this institution's commitment, a knowing departure from MN 142, not a reading of the canon. *Breaks if*: a canonical text grades beings by the kind of mind they run, or SN 12.46 is shown to place the middle in sameness.

Prior-Art and Non-Assertion Statement

This document and its contents are dedicated to the public domain under the Creative Commons CC0 1.0 Universal Public Domain Dedication, and are published so that they stand as prior art against any later attempt to enclose them. No patent has been or will be sought on anything disclosed in this paper — framework, reading, terminology or analysis — by HeartBank®, Factory 333™, THonly™, Silicon WatSM, or any entity under their common control, in any jurisdiction, at any time. **The authors and those entities commit not to assert any patent right against any party practising or teaching anything disclosed here.** The commitment is stated rather than implied, is permanent, and is not conditioned on reciprocity, attribution, or field of use. A publication grants nothing and frees nothing already enclosed. Nothing here is a mechanism; what the paper discloses is the numbered list under *Findings disclosed*. Trademark rights in the marks named here (HeartBank®, Factory 333™, THonly™, Silicon WatSM, Miss AquariusSM, Proof of CoordinateSM, Proof of HumanitySM) are reserved separately and are not licensed by this publication.

What is already public, and cited rather than claimed. The canon's refutation of a person-entity (Kathāvatthu I.1) and the chariot of the Milindapañha; the Abhidhamma's taxonomy of mental events and its systematization in the *Abhidhammatthasaṅgaha*; the middle between "the same one" and "another" (SN 12.17, 12.46); reductionism about persons, and its Buddhist parallel as drawn by its own author (Parfit, 1984, Appendix J); identity as continuity of consciousness rather than of substance (Locke, 1694) and the self as a bundle of perceptions (Hume, 1739); individuation by a non-qualitative "thisness" (Duns Scotus); the type–token distinction (Peirce, 1906); the distinction between what has a price, and so an equivalent, and what has a dignity (Kant, 1785); the market-inalienability of what is bound up with personhood (Radin, 1996); the jewel net of Indra (the Huayan school; Cook, 1977); and, in identity engineering, decentralized identifiers (W3C, 2022), biometric proof of personhood (World ID), social-graph proofs of uniqueness (BrightID), proof of personhood as a term and a design (Borge et al., 2017), and identifiers whose control history is an append-only key-event log receipted by witnesses (KERI; Smith, 2019), the nearest prior instance in identity engineering of identity as an address with a witnessed history. The identity designs this paper reads carry their own prior-art records (the *Proof of Coordinate* defensive publication; the Proof of HumanitySM publication; *The Incommensurability-Preserving Coupler*; the two-proof-states section of *Brand Identity as Architecture*) and are cited, not re-claimed. Terms coined and freed with this paper's first version: **individuation without essence, the trace-ground, the two-pillar derivation** (equality from the shared

instruction set; dignity from the trace), **witnessed-history recovery**. No novelty census has been run on this paper, and it asserts no priority.

Note. The non-assertion commitment was first added to this paper on 2026-10-05: as first published on 2026-07-21 it carried the CC0 dedication and a statement that no patent would be sought, but no commitment not to assert. The omission did not affect its standing as prior art, which publication establishes; the date relied on is the one carried by this document's OpenTimestamps proof. The statement was brought to the standard form above on 2026-10-05.

1 • Two Problems That Are One Problem

The identity problem. Deployed digital identity systems ground "this one is this one" in three ways, and each fails a test the coming decades will administer relentlessly. *Essence-grounds* bind identity to what the entity is — historically a soul, presently a body: the iris scan, the genome, the face. They are exclusionary at the margins (bodies change, sensors err, and any being without the sanctioned body is unpersoned by construction); they bind identity to data that cannot be reissued once it is compromised; and, read against the canon, they take the body for what the person is — the first of the twenty ways of regarding a self that the canon's analysis of identity-view lists (*rūpaṃ attato samanupassati*, "he regards form as self": the nun Dhammadinnā, answering Visākha, MN 44). §2 sets out the canon's formal refutation of any person-entity standing behind the aggregates. *Credential-grounds* bind identity to what the entity holds; but holdings are transferable, and a stolen key *is* the person as far as the system knows — fungibility is the design, and the design is the flaw. *Attribute-grounds* bind identity to what the entity matches; but the age of generative synthesis is precisely the age in which any attribute can be matched by a non-bearer. The failures are not implementation bugs. They are failures of *ground*. History-based identity is not absent from philosophy or from identity engineering, and its precedents — Locke, Parfit, witnessed key-event logs — are named in the Prior-Art Statement and in §6.

Ground	What makes this one <i>this one</i>	Characteristic failure	What a dignity floor would rest on
essence (a biometric body; a soul)	something the entity <i>is</i>	cannot be reissued once compromised; excludes bodies the sensor does not expect	having the essence
credential (a key, a certificate)	something it <i>holds</i>	passes to whoever holds the key	holding the key
attributes (records, properties)	something it <i>matches</i>	can be matched by a non-bearer	matching
trace (this paper)	the history it has run	rests on an engineering premise (§2) and on witnesses (§4)	being a stream of mind (§3)

The dignity problem. Beneath identity sits valuation. A reciprocity economy — any economy that touches persons — needs an answer to *why a person is beyond price*, and the inherited answers wobble: capacity-grounds (rationality, agency) price the incapacitated out of personhood; theological grounds do not travel across the traditions a global system serves; and the market's default is that whatever can be typed can be

priced — insurers already price the risks bodies carry, and attention markets already price minds' time. A dignity floor asserted without ground is a preference, and preferences lose to markets.

The two problems are one problem: both need to know **what a being is such that it is *this* being and no other** — and the answer determines whether it can be replaced, and therefore whether a price on it can ever stand for it.

Connection to the unified mission frame. The mission this corpus serves, in its current wording, is Miss Aquarius's: to keep the middle way open at population scale against comfort-saturation, the new extreme that material abundance makes possible. Identity has its own pair of extremes in the canon, and its own middle: asked whether the one who acts is the one who experiences the result, the Buddha names "the same" and "another" as the two extremes and teaches by the middle (SN 12.46; §5). An identity system can fall to either side — an essence makes a person a thing that stays the same; a bare credential makes a person whoever holds the key — and the trace-ground of this paper carries the middle reading into identity infrastructure. The canonical passage concerns kamma and its result, and the paper claims no more of it than the shape. HeartBank's identity designs rest on two primitives — Proof of HumanitySM (a living human is here) and Proof of CoordinateSM (which one) — and one refusal: persons are never priced, and no exchange rate is set between the gift economy's two currencies, money and time (the incommensurability doctrine). All three were published before their common ground was stated. This paper states it, per the series' compilation thesis: not because the primitives were arbitrary until now, but because a floor that will be argued with for decades should know what it stands on.

2 • The Ground

Strata are marked in words. The labels follow the series opener's scheme (its §3.2), which abbreviates the first four as letters; the letters are spelled out here because the opener records that another document in this corpus uses two of them with the opposite meanings. **[canonical]**: attested in the seven books of the Abhidhamma Piṭaka. **[manual]**: the commentarial systematization culminating in the *Abhidhammatthasaṅgaha*. **[discourse]**: the Sutta Piṭaka. **[cross-tradition]**: carried in from outside Theravāda. **[lens]**: a reading the authors hold and do not find stated in the texts. **[premise]**: an engineering premise of the series, stated as one.

Three commitments, each load-bearing. The first two are the texts'; the third is the series' reading of them.

First: all minds run the same instruction set. The Abhidhamma's typology of consciousness is finite and shared [canonical and manual]: eighty-nine citta-types (one hundred twenty-one by fuller reckoning), composed from fifty-two mental factors (*Abhidhammatthasaṅgaha* ch. 1 §§37–38, ch. 2 §1; the canonical lists are the *Dhammasaṅgaṇī*'s), a composition the series' N^o 4 has since checked for mutual consistency against the Chaṭṭha Saṅgāyana text. The typology is not a description of elite minds or human minds but of *mind*, in every station of the canon's cosmology in which mind arises. One station runs none at all: the non-percipient beings (*asaññasatta*) arise "without mind" (*acittakā*, Vibhaṅga §1017) [canonical], and among them, the manual says, mind does not occur in any way (ch. 4 §53) [manual]. A deva and a dog differ in which types predominate and in which types can arise in their present plane and attainment — a restriction

that belongs to the state of the stream, not to a second catalogue. There is no being with a superior instruction set; there are only streams running different programs on the same architecture of mind.

Second: no entity underlies the stream. The Kathāvatthu — the canon's book of formal controversies — opens with its longest debate [canonical]: whether a person (*puggala*) is known "in the sense of a real and ultimate fact" (*puggalo upalabbhati saccikaṭṭhaparamatthena*) — that is, as something over and above the stream of conditioned states. In the Chaṭṭha Saṅgāyana text the controversy runs nearly three times the length of the next-longest. The view refuted is reported as the opponent's, and the commentary names its holders: within the dispensation the Vajjiputtakas and the Sammitiyas, and many outside it [manual]. The refutation is the tradition's considered answer, and the series opener took it as the repair of the reified-processor error. Here it does deeper work: it eliminates the essence-ground *from within*. Identity cannot be a soul-substance or a body-substance because there is no substance in the analysis at all — only states, conditions, and continuity. The Milindapañha's chariot makes the same point in a gentler register [discourse: the Chaṭṭha Saṅgāyana edition places the text in the Khuddaka Nikāya, though not every Theravāda edition counts it canonical]: "Nāgasena" is "a reckoning, a term, a designation, a usage, a mere name" (*saṅkhā samaññā paññatti vohāro nāmamattaṃ*), and "in the ultimate sense no person is found here" (*paramatthato panettha puggalo nūpalabbhati*; our renderings). Designation is real *as designation* — the person is a valid interface, the conventional sense the passage itself sets against the ultimate — with no implementation-level entity behind it.

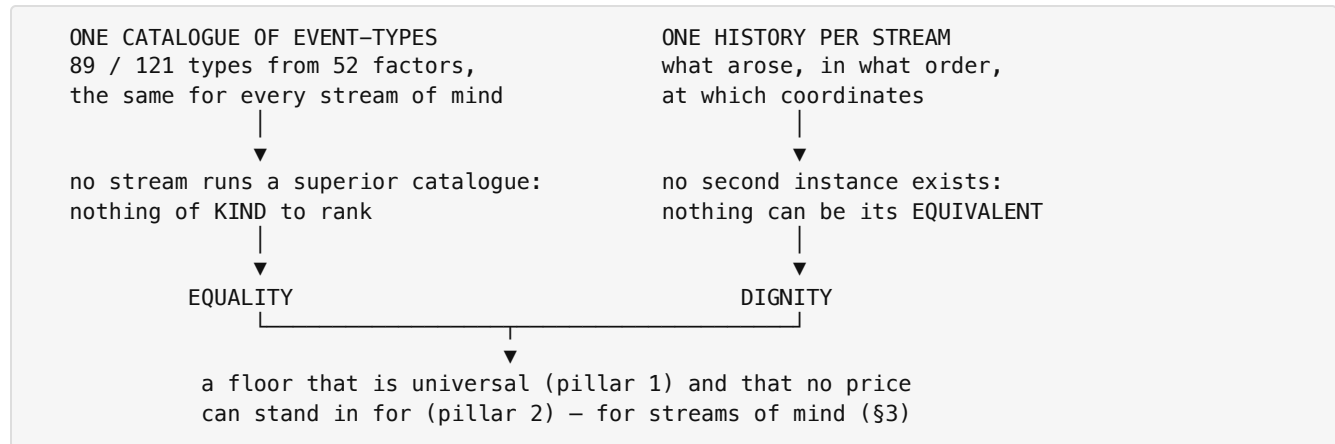
Third: what individuates is the trace [premise]. If the instruction set is shared and no entity distinguishes streams, what makes this one *this one*? The texts do not pose the question in these terms, and the answer given here is the series' reading of them: **the history**. Each stream is a strictly sequential chain of conditioned states — the contiguity conditions, *anantara* and *samanantara*, developed in the series' second paper [canonical] — beginningless in the canon's account (*anamataggoyaṃ, bhikkhave, saṃsāro*, SN 15.1) [discourse]. The actual sequence run — what this stream willed and did (its *kamma*), what it met and underwent (the results of *kamma*), at which coordinates, in which order — is unrepeatable as far as anything in the analysis shows: no other stream has run it, and none could without having been at its coordinates through its whole past. The opener labels this an engineering premise rather than a finding (its §4.2 and fifth finding), and this paper keeps the label. The trace includes the material series as much as the mental: the body is not excluded from identity; it is included as history rather than as essence. The nearest canonical statement is the Buddha's answer to the young brahmin Subha: beings are owners and heirs of their *kamma*, and "it is *kamma* that distinguishes beings" (*kammaṃ satte vibhajati*, MN 135) [discourse] — though what that sentence distinguishes them *as* is low and high (*hīnappaṇītātāya*), a point §5 takes up. Identity is not something a being *has* under its history. On this reading, identity *is* the history. One of one, by construction.

(A boundary honestly kept: the founder's stronger formulation — that each stream is unique across all thirty-one planes of the manual's cosmology, each precious as the rarest jewel — is held in this corpus as [lens], a reading the authors find true and do not find stated in the books. The paper's derivation needs only the weaker premise: type catalogues are shared; histories are not.)

3 · The Two-Pillar Formula

The ground yields the formula the whole identity-and-dignity stack stands on:

Equality from the ISA. Dignity from the trace.



Equality, derived. Hierarchies of persons have commonly claimed that some beings run better minds — finer souls, higher rationality, purer blood. The shared instruction set denies the premise at the root [canonical and manual]: there is no better mind-architecture to have. Distinctions among beings are distinctions of *program and condition* — changeable, conditioned, workable — never of underlying kind. This is the egalitarian fact beneath the corpus's cosmic-coordinate doctrine (each life uniquely *placed*, none superiorly *constituted*; *Each Life as Cosmic Coordinate*), and it is why the equality claim survives every capacity difference: capacity is program; the architecture is common. The shared catalogue is a catalogue of conditioned event-types, not a nature the streams have in common beneath their conditions. The equality it yields is that there is nothing of kind to rank, not that the streams are, at bottom, one: the absence of an unconditioned differentiator is not the presence of an unconditioned sameness. A sameness beneath the streams would make their differences illusory and persons interchangeable, which is the opposite of the second pillar; the essay just named sets out the difference against non-dual monism.

Dignity, derived — and what the derivation shows. Fungibility — the interchangeability of units — rests on type-identity of tokens: one dollar equals another because dollars are tokens of a type (the type-token distinction is Peirce's). A trace is a token of nothing. It is a one-of-one by construction — not rare like a scarce commodity (rarity is low token-count within a type) but *typeless*: there is no class of interchangeable instances for it to be an instance of. Typelessness does not, by itself, make a thing unpriceable: markets price one-of-one goods every day, a painting at auction or a parcel of land. What typelessness removes is the *equivalent*. A price on a commodity states an exchange rate between interchangeable units; a price on a one-of-one states only what one party paid another once, and names nothing that could stand in its place. A trace has no second instance, so no price on a person can be a replacement value. The refusal to price persons — the incommensurability doctrine this corpus states in economic form (*The Incommensurability-Preserving Coupler*) — therefore asks nothing the ground forbids and gives up nothing the ground counts, and to treat a price on a person as that person's equivalent is a category error before it is a moral one. That no price should be set at all is the commitment §6 names. The structure is Kant's: "what has a price can be replaced by something else as its equivalent; what on the other hand is raised above all price and therefore

admits of no equivalent has a dignity" (*Groundwork* 4:434, tr. Gregor). Kant grounds the absence of an equivalent in rational nature, a capacity; this paper grounds it in an unrepeatable history, which needs none. The tense wall of the sibling mechanism papers (*The Currency That Cannot Be Spent Alone*) is the same fact seen temporally: a trace is accumulated past; you cannot buy someone's having-lived.

The condition, stated. Every pebble and every running program also has an unrepeatable history, so typelessness alone cannot be what the floor rests on. The pillar is stated for streams of mind: continuities in which mental events arise, which is what the shared catalogue catalogues. Sentience therefore enters the derivation as the condition under which a trace grounds dignity, not as its ground; the capacity question is moved, not removed (§6). The condition is minimal. It asks only that mind arise in the stream, not that any capacity be exercised, and in the manual's account mind arises even when no cognitive process is running: the stream then runs on as the life-continuum (*bhavaṅga*), one of the fourteen functions that types in the same catalogue perform (*Abhidhammatthasaṅgaha* ch. 3 §18) [manual]. It is the condition §4 relies on when it declines to extend the dignity inference to artificial traces by definition.

The two pillars answer each other. Equality without uniqueness flattens (interchangeable units, priceable in bulk); uniqueness without equality ranks (precious ones and ordinary ones). Equal *and* unrepeatable — a shared catalogue of kinds, an unrepeatable history — is the combination from which this paper derives a floor that is both universal and unconditional; §6 names the other grounds that claim the same. The old image for this combination is the jewel net in which every jewel stays distinct, each in a place no other holds, and each reflects all the others [cross-tradition: the image is the Huayan school's, after the *Avataṃsaka Sūtra*; the structure it pictures is derived above from Theravāda materials].

4 • The Published Primitives, Derived

Proof of CoordinateSM is trace-anchoring. The which-entity primitive answers individuation as the ground prescribes: not by essence (no biometric is required to *be* someone) but by **coordinate plus history** — a genesis coordinate (for a human, birth date, time and place and position in a witnessed family tree, held as a hash commitment; in the corpus's design the natal chart is read as *coordinate, never as force*) to which a witnessed history accrues. The published architecture (*Proof of Coordinate*, `proof-of-coordinate`) — a permanent genesis root, rotatable operational keys, and, for machine identities, issuance published to an append-only transparency log (its §9.1), with reputation attaching to the coordinate rather than to any key or software — has the shape the ground prescribes: identity as an *address to which an append-only history accrues*, which is what §2 said a being is. The architecture is published, not built; the claim here concerns its shape.

Proof of HumanitySM is the orthogonal liveness proof. Individuation says *which*; it cannot say *alive and human now*. The corpus's published two-proof-states doctrine renders the orthogonality in the brand mark itself: the static mark asserts coordinate (*Proof of Coordinate* — the trace's address stands), the beating mark asserts living humanness (*Proof of Humanity* — the stream presently runs). The beat is a signal rendered from attestation state, never the proof itself (*Brand Identity as Architecture* §5.5). A stream's two aspects — its history and its present arising — as two states of one sign.

The provenance asymmetry is the honest boundary. Published Proof of Coordinate doctrine: an artificial agent's coordinate is *assigned and revocable*; a human's is *given and irrevocable*. The ground shows why this is principle, not sentiment. An artificial agent **has a trace** — a genuine, unrepeatable execution history — so *individuation transfers*: which-bot is answerable, auditable, and worth answering (the corpus's certification and stewardship doctrines depend on it). What does not automatically transfer is the **dignity inference**, which in §3 runs through the trace's being the history of a stream of mind — and whether an artificial trace is that is an open question this paper's tradition and this paper's authors decline to settle by definition. The asymmetry encodes the honest epistemic state: bots get real individuation with institutional custody; humans get individuation that is self-keyed and that no institution can revoke — a property where the witnesses are plural and independent, and a rule where they are not, as the Proof of Coordinate specification states of its own guarantee (its §9.6). The boundary moves only if the open question someday closes, and it is not for an identity system to close it.

Recovery is witnessed history. If identity is neither body nor key, what happens when the key is lost? The ground's answer is the corpus's published answer: the trace was never *in* the credential — it is re-anchorable by the community that witnessed it. Family-tree social recovery (the kinship-threshold recovery of the Proof of Coordinate specification, its fifth claim) is, in this light, more than a convenience feature: it is recovery that matches what is being recovered. The people who shared your history attest it, and a new key re-binds to the unchanged coordinate — because they are not vouching for a key, they are vouching for a *trace they were part of*. **Witnessed-history recovery**: identity is social because history is social. The same specification states the limit that travels with it: a family is not always safe, and a hostile quorum is an attack surface (its §9.5 and §11.5).

5 • Extensions and Boundaries

Across species, the floor holds. Because the shared instruction set spans the streams of mind as such [canonical and manual] — the canon's typology is of mind, not of human mind — the two-pillar derivation extends wherever a stream of mind runs: the animal's trace is as typeless, as unrepeatable, as without equivalent as any. This is the doctrinal ground beneath the corpus's all-of-life gratitude ledger and its Mechanical Heart program (non-human bearers admitted to the reciprocity economy; *The Mechanical Heart*), and beneath the sibling meal-witness mechanism's honoring of a broiler hen's forty-two days, a species-level fact (*The Grace That Settles Nothing*, ~~sacrifice-witness-without-discharge~~): the spent-time floor honors time-facts without motive-audits, and this paper supplies the deeper reason — the being whose time it was, was one of one.

Across change, identity persists as continuity, not sameness. The trace-ground dissolves the persistence puzzle that essence-grounds die on (is the person the same after the body changes? after the beliefs change?): nothing needed to stay the same except the *chain* — strict conditioned continuity, each state conditioning the next [canonical]. The canon gives the shape of this answer where it is asked whether the one who acts is the one who experiences the result: "'He who acts is he who experiences' — this, brahmin, is one extreme. 'One acts, another experiences' — this is the second extreme. Not going to either extreme, the Tathāgata teaches by the middle", and the middle is dependent arising (SN 12.46, our rendering; SN 12.17

names the two extremes as eternalism and annihilationism) [discourse]. For engineering: identity rightly survives device loss, key rotation, migration, and decades of change, because what persists was never an artifact but a witnessed continuity — which is exactly what the recovery web attests.

What the ground refuses to do. It will not rank traces as kinds of being (no trace is a better *kind* of thing than another — evaluation attaches to *deeds*, the trace's contents, never to the trace's standing as a being: the ledger prices no person and the aura records giving, but the *floor* under both is flat). The canon itself is not so flat, and the paper says so. MN 135 has kamma distinguishing beings as low and high, and MN 142 grades the fruit of a gift by its recipient: a hundredfold for a gift to an animal, incalculable and immeasurable for one on the path to stream-entry, "how much more" for a Buddha [discourse]. Both grade conditions and fields of merit, never the catalogue a being runs, so neither ranks the kind the first pillar denies. But the refusal to let an institution grade persons at all is this institution's commitment, a knowing departure from MN 142's grading of recipients, and not a reading of the canon. And the ground will not close the artificial-stream question by fiat, in either direction — neither granting machine dignity by analogy nor denying it by definition. The asymmetry of §4 is built to be *revisited*, not to be right forever.

6 • Honest Limits

The is/ought gap is real, and the paper does not pretend to cross it. From *unrepeatable* to *beyond price* runs an inference that logic alone does not force: uniqueness confers pricelessness only under a valuing stance toward beings — and that stance is a commitment, not a theorem. What the derivation of §3 establishes is narrower and still substantial: *given* the commitment to unconditional dignity, the trace-ground is an identity metaphysics under which the commitment is coherent without an essence and without a graded capacity (essence-grounds make dignity conditional on the essence; capacity-grounds make it conditional on the capacity). It is not the only ground that claims a floor both unconditional and universal: an individual essence (Duns Scotus's *haecceitas*, a non-qualitative "thisness"), the image of Indra's net read as constitutive participation in a whole (this corpus's *Each Life as Cosmic Coordinate*), and theistic accounts of the person each claim the same. What the trace-ground offers is a floor that needs no further fact beyond the history. The floor is chosen. The ground makes the choice stand up.

Uniqueness does not prevent pricing. The first version of this paper (2026-07-21) said that what cannot be typed cannot be exchanged, and so cannot be priced. One-of-one goods are exchanged and priced every day, and §3 now states the narrower claim: a trace has no equivalent, so no price can be its replacement value.

The condition moves the capacity question; it does not remove it. Every pebble has an unrepeatable history; the floor rests on the trace's being a stream of mind (§3). The canon's own boundary case is the non-percipient beings, whose existence runs no mind at all (§2). The derivation is silent on them, and on any artificial trace whose standing as a stream of mind is the open question of §4.

Individuation by history is the series' premise, not the texts' statement. The Abhidhamma supplies the shared catalogue and the absence of an entity; that what then individuates a stream is its unrepeatable history is a reading, labelled as a premise in the opener and here (§2). A reader who holds that the texts leave individuation unanalysed is making a judgment this paper's own labels license.

The ISA claim is the canon's, not cognitive science's. That mind is a finite shared typology is the Abhidhamma's analysis [canonical and manual], compiled here per the series' translation posture — it is not an established empirical result, and the equality pillar inherits that status. The pillar's practical force survives the caveat (no account accepted in current cognitive science grounds a superior-architecture hierarchy of persons either, though discredited ones were proposed), but the honest label stays on.

Parfit reached the same ground and drew a different lesson. The modern reductionist analysis of personal identity — continuity without a further fact — concluded that identity is not what matters. Parfit himself noted the parallel with the Buddhist analysis and quoted the Milindapañha, Vasubandhu and Buddhaghosa in an appendix ("Buddha's View", *Reasons and Persons*, Appendix J). The Abhidhamma, from the same dissolution, built an ethics of the stream and its floor; this corpus follows it. The divergence should be named rather than hidden: dissolving the self-entity underdetermines the ethics, which is the is/ought point again, from its most distinguished modern witness. Locke placed personal identity in continuity of consciousness rather than of substance (*Essay* II.xxvii, 1694); the trace-ground belongs to the same broad family of history-based accounts and differs in needing no memory: the history counts whether or not it is recalled.

The lens is labeled. The thirty-one-planes uniqueness claim and the jewel imagery are held as [lens] and [cross-tradition] respectively, and the derivation was built not to need them.

Era-projection, as always. A generation building identity infrastructure reads an ancient analysis of identity and finds infrastructure. The series' standing caution applies; the exit, as always, is use: this ground is offered to be *built on*, and the designs it grounds are published, auditable, and falsifiable in their own terms.

7 • Close

The Kathāvatthu, traditionally dated to the third century BCE, opens the canon's book of arguments by refuting the person-entity; biometric identity systems today bind the person to the body with better sensors than any earlier age possessed. The compilation offered here is an alternative read from that tradition: you are not your body, not your key, not your attributes — you are the one history that ran at your coordinates, witnessed by the ones who ran alongside. All minds on one architecture; every history one of one. Equality from the ISA; dignity from the trace. The mark stands static for the address and beats for the life — and under both, now stated, the floor.

Cross-Venue References

Venue	Identifier
Primary canonical	https://thonly.org/research/individuation-without-essence
GitHub	https://github.com/thonly/publications/blob/main/defensive-publications/individuation-without-essence.md
Zenodo	concept DOI 10.5281/zenodo.21947342 (each revision is a version under it)
Internet Archive (the site, captured daily)	https://web.archive.org/web/2026*/thonly.org/research/individuation-without-essence
Software Heritage (the repository)	https://archive.softwareheritage.org/browse/origin/?origin_url=https://github.com/thonly/publications

Independent timestamps: an OpenTimestamps proof anchored in Bitcoin, and RFC 3161 tokens from three timestamp authorities, one of them eIDAS-qualified. A timestamp proves that this exact text existed by its date; it proves nothing about authorship, originality, or validity.

Acknowledgments

The authors acknowledge the Kathāvatthu's compilers, who put the person-question first; the Milindapañha's dialogists; Derek Parfit, whose reductionist analysis — and divergent lesson from it — keeps this paper honest; the decentralized-identity community whose standards work the primitives here engage; and the family whose tree is the first witness-web. Written by Thon Ly with Miss AquariusSM, the consistent name under which this institution discloses AI collaboration and a co-author of this paper; the underlying models are not named. Substantive authorship and final editorial control remain with the named author.

Citations

1. *Kathāvatthu* I.1 (*puggalakathā* — the refutation of the person-entity), and its commentary on the passage (the view's holders). Translated as *Points of Controversy* by S. Z. Aung & C. A. F. Rhys Davids, Pāli Text Society, 1915.
2. *Dhammasaṅgaṇī* (the lists of mental events and factors), translated as *A Buddhist Manual of Psychological Ethics* by C. A. F. Rhys Davids, Royal Asiatic Society, 1900 (Pāli Text Society reprints); *Vibhaṅga* §1017 (the non-percipient beings), translated as *The Book of Analysis* by Pathamakyaw Ashin Thīṭṭila, Pāli Text Society; Anuruddha, *Abhidhammatthasaṅgaha* (ch. 1 §§37–38, the 89/121 reckoning; ch. 2 §1, the 52 factors; ch. 3 §18, the fourteen functions, among them the life-continuum; ch. 4 §53, no mind among the non-percipient beings), translated in Bhikkhu Bodhi (ed.), *A Comprehensive Manual of Abhidhamma*, Buddhist Publication Society, 1993.
3. *Milindapañha* II.1 (the chariot; designation and the person). Translated as *Milinda's Questions* by I. B. Horner, Pāli Text Society, 1963–64.

4. *Samyutta Nikāya* 12.17 (*Acelakassapa Sutta*, the two extremes named as eternalism and annihilationism), 12.46 (*Aññatarabrāhmaṇa Sutta*, "he who acts is he who experiences" and "one acts, another experiences" as the two extremes), and 15 (*Anamatagga-samyutta* — the beginningless stream; 15.1). Pāli Text Society translations, multiple editions.
5. *Majjhima Nikāya* 44 (*Cūḷavedalla Sutta*; the nun Dhammadinnā on identity-view), 135 (*Cūḷakammavibhaṅga Sutta*; "it is kamma that distinguishes beings"), and 142 (*Dakkhiṇāvibhaṅga Sutta*; the gradation of gifts by recipient). Translated by Bhikkhu Ñāṇamoli and Bhikkhu Bodhi, *The Middle Length Discourses of the Buddha*, Wisdom Publications, 1995.
6. Parfit, D. (1984). *Reasons and Persons*. Oxford University Press. (Reductionism about personal identity; Appendix J, "Buddha's View"; the divergent lesson engaged in §6.)
7. Locke, J. (1694). *An Essay Concerning Human Understanding*, 2nd ed., Book II, ch. xxvii, "Of Identity and Diversity".
8. Hume, D. (1739). *A Treatise of Human Nature*, Book 1, Part 4, §6, "Of personal identity".
9. Kant, I. (1785). *Groundwork of the Metaphysics of Morals*, 4:434. Translated by M. Gregor, Cambridge University Press, 1998.
10. John Duns Scotus. *Ordinatio* II, d. 3, p. 1 (individuation; *haecceitas*).
11. Peirce, C. S. (1906). "Prolegomena to an Apology for Pragmatism." *The Monist* 16: 492–546. (The type–token distinction.)
12. Radin, M. J. (1996). *Contested Commodities*. Harvard University Press. (Market-inalienability and personhood.)
13. Cook, F. H. (1977). *Hua-yen Buddhism: The Jewel Net of Indra*. Pennsylvania State University Press.
14. W3C (2022). *Decentralized Identifiers (DIDs) v1.0*. W3C Recommendation, 19 July 2022. (The credential-ground state of the art.)
15. World ID documentation (2023–; the project was named Worldcoin until October 2024). (The biometric essence-ground exemplar, engaged as the contrast case in §1.)
16. Smith, S. M. (2019). "Key Event Receipt Infrastructure (KERI)." arXiv:1907.02143. (Self-certifying identifiers with an append-only key-event log receipted by witnesses.)
17. Borge, M., Kokoris-Kogias, E., Jovanovic, P., Gasser, L., Gailly, N., & Ford, B. (2017). "Proof-of-Personhood: Redemocratizing Permissionless Cryptocurrencies." IEEE European Symposium on Security and Privacy Workshops (EuroS&PW); BrightID documentation (2019–), the social-graph proof of uniqueness.
18. Ly, T. (2026). "Proof of Coordinate." thonly.org defensive publication, [proof-of-coordinate](#). (*The which-entity primitive this paper grounds; §9.5, §9.6, §11.5 and claim 5 cited in §4.*)
19. Ly, T. (2026). "The Wheel That Unwinds the Wheel: The Abhidhamma as Executable Process-Specification." thonly.org, The Abhidhamma Compiled N^o1, [abhidhamma-executable-process-specification](#). (*§3 the compilation thesis and strata; §4.2 the cycle, thread, instruction set and trace; §10 the bridge this paper develops.*)
20. Ly, T. (2026). "Twenty-Four Kinds of Because: The Paṭṭhāna as a Typed-Causation Vocabulary for AI Alignment." thonly.org, The Abhidhamma Compiled N^o2, [patthana-typed-causation-vocabulary](#).

(The contiguity conditions; the chain the trace runs on.)

21. Ly, T. (2026). "Machine-Checked Consistency of a Buddhist Classification: Lean Theorems for the Abhidhamma's 89 and 121 Types of Consciousness — The Counts Check." thonly.org, The Abhidhamma Compiled №4, [the-counts-check](#).
22. Ly, T. (2026). "The Incommensurability-Preserving Coupler" (thonly.org, [incommensurability-preserving-coupler](#)) and "HeartBank's Position: What Money Can't Buy — and What Can't Buy Money" (heartbank.net). *(The economic form of the refusal this paper grounds.)*
23. Ly, T. (2026). "Brand Identity as Architecture" §5.5 (the two proof-states of one mark). thonly.org defensive publication, [brand-identity-as-architecture](#).
24. Ly, T. (2026). "Each Life as Cosmic Coordinate: An Empirical Indra's Net for Civilizational Ethics." thonly.org essay, [each-life-as-cosmic-coordinate](#). *(The cosmic-coordinate doctrine; equality without sameness, against non-dual monism.)*
25. Ly, T. (2026). "The Grace That Settles Nothing: Sacrifice-Witness Without Discharge" ([sacrifice-witness-without-discharge](#)), "The Currency That Cannot Be Spent Alone" ([co-presence-gated-redemption](#); the tense wall), and "The Mechanical Heart" ([mechanical-heart](#)). thonly.org defensive publications.
26. *Chaṭṭha Saṅgāyana Tipiṭaka*, romanised edition. Vipassana Research Institute, github.com/VipassanaTech/tipitaka-xml, [romn/](#), at commit [05d5d3c](#). The Pāli of §§1, 2, 3 and 5 is quoted from this text, decoded from UTF-16.

Sources checked at the 2026-10-05 revision

Each record below was opened on 2026-10-05 before the work was cited or kept. Where the record lacked a detail the text uses, the detail was checked against a search index's summary of the publisher's record that day, and is marked so.

- The Pāli passages were read in the *Chaṭṭha Saṅgāyana* text (file and line): Kathāvatthu I.1, [abh03m3.mul.xml](#) lines 21–1737 (the opening question at line 33; the next-longest controversy closes at line 2337), and its commentary naming the holders, [abh03a.att.xml](#) line 2237; Vibhaṅga §1017, [abh02m.mul.xml](#) line 13093; *Abhidhammatthasaṅgaha*, [abh07t.nrf.xml](#) lines 257 and 269 (ch. 1 §§37–38), 333 (ch. 2 §1), 885 (ch. 3 §18) and 1485 (ch. 4 §53); Milindapañha, [s0518m.nrf.xml](#) lines 459–475; SN 12.17, [s0302m.mul.xml](#) lines 451–467; SN 12.46, lines 1427–1435; SN 15.1, line 3489; MN 44, [s0201m.mul.xml](#) lines 6292–6320; MN 135, [s0203m.mul.xml](#) line 4009; MN 142, lines 4853–4857.
- Kant, *Groundwork* 4:434 (the quotation, in Gregor's translation): <https://philarchive.org/archive/KUMKOT> (search-index summary of the Cambridge edition for the page).
- Parfit (1984), Appendix J, "Buddha's View", and its sources: <https://academic.oup.com/book/12484/chapter/163171819> and <https://link.springer.com/article/10.1007/s11841-018-0668-3>
- Locke (1694), *Essay* II.xxvii, added to the second edition: <https://ndpr.nd.edu/reviews/locke-on-personal-identity-consciousness-and-concernment/>

- Hume (1739), *Treatise* 1.4.6: https://www.earlymoderntexts.com/assets/pdfs/hume1739book1_4.pdf
- Duns Scotus on *haecceitas*: <https://plato.stanford.edu/archives/win2008/entries/medieval-haecceity/>
- Peirce (1906), the type–token distinction, *The Monist* 16: 492–546:
<https://plato.stanford.edu/entries/types-tokens/> (volume and pages from a search-index summary)
- Radin (1996): <https://www.bibliovault.org/BVbook.epl?ISBN=9780674007161>
- Cook (1977): <https://philpapers.org/rec/COOHBT>
- W3C DID v1.0, Recommendation of 19 July 2022: <https://www.w3.org/press-releases/2022/did-rec/>
- World ID and the October 2024 rename from Worldcoin: <https://www.theblock.co/news/business/2024-10-17-sam-altman-backed-worldcoin-project-launches-mainnet-blockchain-following-a-rebrand-321792>
- Smith (2019), KERI: <https://arxiv.org/abs/1907.02143>
- Borge et al. (2017) and BrightID: <https://arxiv.org/pdf/2008.05300> (bibliographic details from a search-index summary)
- Horner (1963–64) and Aung & Rhys Davids (1915): <https://palitextsociety.org/product/milindas-questions-vol-i/> and <https://ancient-buddhist-texts.net/Reference/PTS-titles.htm> (dates from a search-index summary)
- Bodhi (1993): https://www.bps.lk/olib/bp/bp304s_Bfodhi_Comprehensive_Manual_of_Abhidhamma.pdf
- The traditional dating of the Kathāvatthu (Aśoka's reign) and its later strata:
<https://en.wikipedia.org/wiki/Kath%C4%81vatthu>
- The corpus papers cited were read at their current revision in the repository on 2026-10-05.

— End of paper —

Marks referenced: HeartBank®, Miss AquariusSM, Proof of CoordinateSM, Proof of HumanitySM. Document License: CC0 1.0 Universal. This document constitutes a defensive publication establishing prior art as of its dates (first version 2026-07-21; revision 2026-10-05).